

Lab Assignment 4: Schema Design

Group Name: The DB Architects

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Database Schema Cards

Legend: **Blue** (PK) = Primary Key, **Orange** (FK) = Foreign Key, **Purple** (PK, FK) = Composite/Inherited Key.

Users (Base Table)	
user_id (PK)	Integer
name	Varchar(100)
email	Varchar(100)
password	Varchar(255)
phone_number	Varchar(20)
country	Varchar(50)
gender	Varchar(10)
role_type	Enum

Universities	
university_id (PK)	Integer
name	Varchar(255)

Programs	
name (PK)	Varchar(50)

Students	
user_id (PK, FK)	Integer
student_roll_id	Varchar(20)
date_of_birth	Date
cgpa	Decimal(3,2)

Instructors	
user_id (PK, FK)	Integer
faculty_id	Varchar(20)
post	Varchar(50)
university_id (FK)	Integer

Courses	
course_id (PK)	Integer
name	Varchar(255)
credits	Integer
university_id (FK)	Integer
textbook_isbn (FK)	Varchar(20)

Topics	
topic_id (PK)	Integer
name	Varchar(100)

Taught_By	
instructor_id (PK, FK)	Integer
course_id (PK, FK)	Integer

Has_Content	
course_id (PK, FK)	Integer
content_id (PK, FK)	Integer

Enrolled_In	
student_id (PK, FK)	Integer
course_id (PK, FK)	Integer
marks	Decimal(5,2)

Prerequisites	
course_id (PK, FK)	Integer
prereq_id (PK, FK)	Integer

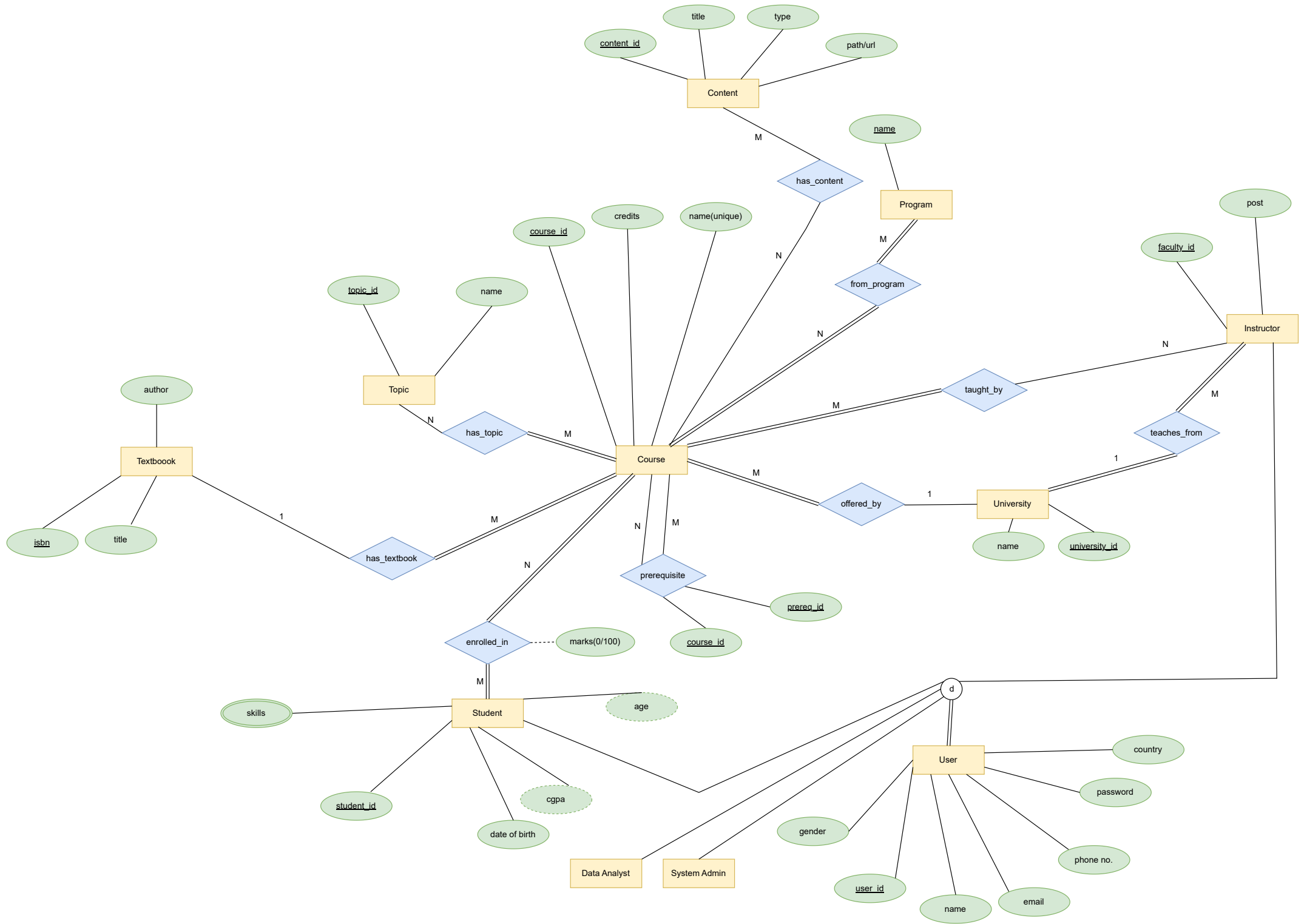
Content	
content_id (PK)	Integer
title	Varchar(255)
url	Text
type	Varchar(20)

Textbooks	
isbn (PK)	Varchar(20)
title	Varchar(255)
author	Varchar(255)

Program_Courses	
program_name (PK, FK)	Varchar(50)
course_id (PK, FK)	Integer

Course_Topics	
course_id (PK, FK)	Integer
topic_id (PK, FK)	Integer

Student_Skills	
user_id (PK, FK)	Integer
skill_name (PK)	Varchar(50)



System Architecture Specification

Web-Based Information System Implementation

1 Technology Stack & Architecture

To meet the requirement for a “complete web-based information system,” our group has selected a stack that ensures rapid development and robust database connectivity.



Figure 1: 3-Tier Web Architecture Implementation

1.1 Frontend (Client Layer)

- **HTML5 / CSS3 / Bootstrap:** We utilize the Bootstrap framework to build a responsive “Entry Page” that dynamically adapts the menu interface based on the user category (Student, Admin, Instructor, or Analyst).

1.2 Connectivity Server (Application Layer)

- **Python (Flask Framework):** Flask acts as the middleware connectivity server. It intercepts HTTP requests (such as “Register for Course”), handles session management for the four distinct user roles, and securely executes SQL commands.
- **Jinja2 Templating:** This engine is used to dynamically render database content—such as lists of available courses or student marks—directly into the HTML views.

1.3 Backend (Database)

- **MySQL:** A relational database management system selected to store the complex relationships between Users, Courses, and Partner Universities as defined in our schema.

2 Technical Flow (Request Lifecycle)

The system follows a standard 3-Tier Web Architecture. The flow of data for a typical operation (e.g., an Instructor adding content) is as follows:

Request

The user interacts with the frontend interface (e.g., filling out a form to “Add Content”) and submits a `POST` request via the browser.

Processing

The Flask server receives the request. It first validates the user’s session to ensure they have the correct “Instructor” role and sanitizes the input data.

Database Interaction

Flask executes the corresponding SQL command (e.g., an `INSERT` query) into the `Content` table in the MySQL database.

Response

Upon successful execution, the database returns a status signal. Flask then re-renders the specific Course Page to display the newly added material to the user.